

Anshul Sharma

Data Analyst / Data Scientist / ML Engineer



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Portfolio: <https://web-anshul.vercel.app/>

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GitHub: <https://github.com/anshul-1999/>

Professional summary

Data Scientist and Data Analyst with 2.5+ years of experience in data analysis, machine learning, and scalable data systems. Skilled in building automated data pipelines, implementing distributed computing systems, and developing end-to-end ML solutions. Currently pursuing an M.Sc. in Data Science in Germany. Strong background in Python, SQL, and system-level problem solving.

Core competencies

- Programming: Python, SQL, C++, Java (Basic)
- Data Science & ML: Machine Learning, Clustering, TF-IDF, Feature Engineering, Model Evaluation
- Data Engineering: ETL Pipelines, Data Processing, Automation
- Tools & Frameworks: Pandas, NumPy, Flask, FastAPI, FAISS, Git
- Distributed Systems: MPI (Message Passing Interface), Multithreading
- Data Visualization: Tableau (Basic)
- Languages: English (C1), German (B1), Hindi (Native)

Professional Experience

Senior Project Technical Assistant

IIT Bombay, Mumbai, India

Nov 2021 – Jun 2024

- Conducted data analysis and supported development for the Indian Culture Portal project
- Designed and maintained structured data workflows to improve system efficiency
- Applied data analytics and visualization to monitor user engagement and system performance
- Automated data handling processes, improving workflow efficiency and reducing manual effort
- Produced technical documentation and contributed to system architecture decisions

Data Science Intern

Flip Robo Technologies (Remote)

Dec 2022 – June 2023

- Built automated data pipelines for preprocessing and analysis tasks
- Performed data cleaning, transformation, and feature engineering using Python and Pandas
- Applied machine learning models for predictive analysis and clustering tasks
- Improved data processing efficiency through modular and reusable code design

Projects

Crime Data Analysis (Capstone Project) | Python, SQL, Pandas, Tableau, ML

GitHub: <https://github.com/anshul-1999/Crime-Capstone>

- Analyzed large-scale crime datasets using Python, SQL, and Pandas
- Applied clustering algorithms to identify high-risk zones and crime patterns
- Built visual dashboards in Tableau to present insights and trends
- Performed data preprocessing, feature engineering, and exploratory data analysis

Pryas – Bachelor's Thesis

https://www.ijaresm.com/uploaded_files/document_file/Anand_Kumar,_Anshul_Sharma,_Abhishek_Mishra,_Ashish_Kumar_Singh,_Swati_TyagiHJR1.pdf

- Development of a web-based school administration system that integrates online exams, note exchange, and information management functions.

Distributed File Scanning System | C++, MPI, Multithreading, Distributed Systems

GitHub: <https://github.com/anshul-1999/Multi-Node-Distributed-File-Scanning-System-C-MPI>

- Developed a distributed parallel file scanning system using MPI and multithreading
- Designed a Coordinator-Worker architecture for efficient task distribution across nodes
- Implemented hashing (MD5, SHA256) for duplicate detection and file analysis
- Improved scalability and processing speed through parallel execution

Information Retrieval System | Python, Flask, NLP, TF-IDF

GitHub: <https://github.com/anshul-1999/Offline-Information-Retrieval-IR-Search-System-Python-Flask->

- Built a search engine using TF-IDF vectorization and cosine similarity for ranked retrieval
- Developed a Flask-based web interface for real-time keyword search and analytics
- Evaluated system performance using Precision@K metrics
- Applied NLP techniques for efficient indexing and query processing

AI Document Search & Chat (RAG System) | FastAPI, FAISS, LLM, NLP

GitHub: https://github.com/anshul-1999/AI_DOC_Search

- Built a Retrieval-Augmented Generation (RAG) system for querying PDF documents using local LLMs
- Implemented hybrid search combining BM25 and vector embeddings to improve retrieval accuracy
- Developed a FastAPI backend enabling real-time conversational interaction with documents
- Designed context-aware responses with conversational memory for improved user experience

AI Assistant with Memory & Voice | LLM, Whisper, Vector DB, NLP

GitHub: https://github.com/anshul-1999/AI_Assistant

- Built a local AI assistant with voice interaction and persistent memory
- Integrated speech recognition (Whisper) and LLM-based conversational system
- Designed a vector database for long-term memory and context retention
- Implemented filtering mechanisms to improve response relevance and safety

Education

M.Sc. Data Science (Expected 2027)

XU Exponential University, Potsdam, Germany

B.Tech Computer Science and Engineering

Dr. A.P.J. Abdul Kalam University, India

2017 – 2021

Certifications

- BCG Data Science Job Simulation – Forage (2025)
- Quantum Data Analytics Simulation – Forage (2025)
- PG Program in Data Science & Machine Learning – Datatrained (2023)
- Python for Data Science
- SQL for Data Science
- Data Wrangling & Visualization
- Applied Statistics